



Incremental updating reduction for relation decision systems with dynamic conditional relation sets

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ABSTRACT

In real applications, the feature set in a relation decision system often varies with time resulting in a dynamic relation decision system where the existing attribute reduction methods become time-consuming and not suitable. How to efficiently update the reduction with prior information of attribute reduction is an important task. Aim at this, we firstly propose an incremental updating mechanism for the positive region and right neighbor of a relation decision system when the conditional relation set has increased or decreased. By integrating the proposed incremental updating mechanism of positive region and right neighbor to the positive region-based reduction method, a novel incremental updating reduction algorithm for relation decision systems with dynamic conditional relation sets is designed. The proposed incremental updating reduction algorithm can speed up the reduction of relation decision systems with dynamic conditional relation sets. Specially, it can deal with the reduction of dynamic decision systems whose decision relations aren't equivalence relations, such as incomplete decision systems with missing decision values. The experimental study carried out on UCI datasets show the good performance the proposed algorithm.

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1. Introduction

Rough set theory put forward by Pawlak [25,26] is an effective mathematical tool for mining hidden knowledge from uncertain data. Being one of the most important applications of rough sets, attribute reduction plays a vital role in machine learning and pattern recognition, and provides a theoretical framework for feature selection of decision systems with symbols and discrete data. With the progress of study and application of rough set [1,22,33,45], more attention has been paid to attribute reduction methods for decision systems [7,10,14,27,32,38,39,41,49].

To deal with various new decision systems such as set-valued decision systems, different extension models of rough sets have been proposed resulting in different types of attribute reduction methods including distribution reduction [42], variable precision reduction [21], and covering reductions [4,36]. In [13], Kryszkiewicz investigated attribute reduction of consistent and inconsistent incomplete decision systems based on similarity relations. Yang et al. [47] developed attribute reduction algorithm for incomplete ordered information systems based on the dominance-based rough set. More and more attention has been paid to incomplete decision systems [15]. It is worth noting that the above-mentioned attribute reduction

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methods were all proposed for a certain type of decision system. In [37], Wang et al. proposed the concept of relation decision systems, and conducted systematical study on the reduction of relation decision systems where the conditional relations (conditional attributes) are general relations. Although the reduction algorithms for such relation decision systems perform well in various types of decision systems such as complete, incomplete, numerical and covering ones, they show weakness in decision systems with missing decision values where the decision relations are no longer equivalence relations.

Recently, Liu et al. [16–18] extended Wang's research by redefining the relation decision systems where the decision relations are no longer limited to equivalence relations, and established a general discernibility matrix-based reduction algorithm which can find all reductions of a relation decision system. However, this reduction algorithm ignores the computationally complexity. When dealing with the reductions of relation decision systems with large-scale data or dynamic relation decision systems, it may not be suitable or invalid. It should be pointed out that among the reduction algorithms which can be divided into three categories: positive region-based ones [20,24,28], discernibility matrix-based ones [30,43], and entropy-based ones [15,31], the positive region-based algorithms [29] can indeed improve the efficiency of reduction. Thus, the positive region is employed in the attribute reduction algorithms proposed by this paper.

In real-world applications, a decision system usually varies with time in attribute set [9], object set [35] or attribute values [40]. Such a decision system is called a dynamic decision system. In the face of dynamic decision systems, how to get its reduction becomes the primary problem. For this problem, one way to get the reduction after variation is to do from scratch with ignoring the reduction obtained before variation. We call such a reduction algorithm non-incremental reduction algorithm. It can be easily seen that a non-incremental reduction algorithm tends to be time-consuming or even infeasible [48]. Incremental learning, being an efficient method for dealing with dynamic decision systems and mining dynamic data, can avoid some recalculations by using previous data [8]. Hu et al. [11] proposed an incremental approach for updating attribute reduction based on elementary sets, which can be only applied to complete information systems. Based on the positive region reduction method, Shu et al. [34] proposed an incremental updating mechanism for tolerance classes and a dynamic attribute reduction algorithm for incomplete decision systems. For decision systems with dynamic object sets, Chen et al. [5] proposed an incremental method to solve the updating problem of the approximate set based on equivalence relation.

Although the studies on incremental reduction research of dynamic decision systems are fairly progressive [6,23,44,46], the focus of the existing incremental methods are dynamic decision systems whose decision relations are equivalence. When facing with dynamic decision systems whose decision relations are not equivalence ones, the existing reduction methods may be inapplicable or invalid. In fact, these problems can be attributed to the reduction problem of dynamic relation decision systems. In real application, the dynamic decision systems whose changes are caused by the change of attribute set occurs in most cases [12,50]. Therefore, this paper concentrates on the incremental updating reduction algorithms for relation decision systems with dynamic conditional relation sets.

The main contributions of this study are summarized as follows: (1) an incremental updating reduction algorithm is proposed for relation decision systems whose conditional relation sets have increased; (2) an incremental updating reduction algorithm is proposed for relation decision systems whose conditional relation sets have decreased; (3) the proposed algorithms unify the reduction of some dynamic decision systems including those whose decision relations are not equivalence relations; (4) experiments carried out on different data sets show the effectiveness of the proposed algorithms.

The rest of this paper is organized as follows. In Section 2, some important concepts about relation decision systems are reviewed. Based on the positive region-based reduction method, Section 3 presents the incremental updating mechanisms for positive region and right neighbor in relation decision systems with dynamic conditional relation sets. In Section 4, the proposed updating mechanisms of positive region and right neighbor are employed to update the reductions of relation decision systems. We propose the incremental updating reduction algorithms for relation decision systems with dynamic conditional relation sets. In Section 5, numerical experiments are conducted to verify our proposed algorithms. Section 6 gives a conclusion of the paper and discusses the future work.

2. Preliminaries

In this section, we recall some basic definitions and properties of binary relations and relation decision systems [16,37]. Suppose that $U = \{x_1, x_2, \dots, x_n\}$ is a finite set of objects called the universal set and $P(U)$ is the power set of U . Let R be a binary relation on U . The left and right neighborhood of x with respect to R are respectively defined as follows.

$$l_R(x) = \{y \in U : (y, x) \in R\} \text{ and } r_R(x) = \{y \in U : (x, y) \in R\}.$$

Based on the left and right neighborhoods, several special binary relations are given as follows.

- (1) R is referred to as reflexive if $x \in r_R(x)$ for any $x \in U$;
- (2) R is referred to as symmetric if $r_R(x) = l_R(x)$ for any $x \in U$;
- (3) R is referred to as s transitive if $y \in r_R(x) \Rightarrow r_R(y) \subseteq r_R(x)$ for any $x \in U$.

Moreover, for any $X \subseteq U$, $\underline{R}(X) = \{x : r_R(x) \subseteq X\}$ and $\bar{R}(X) = \{x : r_R(x) \cap X \neq \emptyset\}$ are called the lower and upper approximations of X , respectively.

Theorem 2.1 [16,37] Suppose U is a finite universe and R_1, R_2 are two binary relations on U .

Then the following properties hold:

- (1) $R_1 \subseteq R_2 \iff r_{R_1}(x) \subseteq r_{R_2}(x)$ for any $x \in U$;
- (2) $r_{(R_1 \cap R_2)}(x) = r_{R_1}(x) \cap r_{R_2}(x)$ for any $x \in U$;

(3) $r_{(R_1 \cup R_2)}(x) = r_{R_1}(x) \cup r_{R_2}(x)$ for any $x \in U$.

As is well known, a knowledge base is a relation system, and a relation decision system is a decision information system when the relations induced by the condition attributes and the decision attributes are all equivalent relations. Therefore, a relation decision system is a generalization of a decision information system (shown in Fig. 4). In fact, some decision systems, such as covering decision systems, set-valued decision systems, and incomplete decision systems (shown in Figs. 1–3), can be regarded as special relation decision system [16,37].

In order to represent the relationship among objects in a relation decision system, this paper introduces the relation matrix and its operations. Given $U = \{x_1, x_2, \dots, x_n\}$, let $R = \{R_1, R_2, \dots, R_m\}$ be a family of binary relations on U , then for any $R \in R$, the relation matrix of R denoted by $M_R = (c_{ij})_{n \times n}$ is defined as $c_{ij} = \begin{cases} 1, & (x_i, x_j) \in R \\ 0, & (x_i, x_j) \notin R \end{cases}$. For any two $P, Q \in R$, whose relation matrices are $M_P = (p_{ij})_{n \times n}$ and $M_Q = (q_{ij})_{n \times n}$ respectively, $M_{\{P, Q\}} = M_P \wedge M_Q = (c_{ij})_{n \times n}$ is defined as $c_{ij} = p_{ij} \wedge q_{ij}$. Further details can be found in [3,12,19,41].

Example 2.1 shows two relation matrix representations and their operations.

Example 2.1. Let $U = \{x_1, x_2, x_3, x_4, x_5\}$ and $R = \{R_1, R_2\}$, where $R_1 = \{(x_1, x_3), (x_1, x_4), (x_2, x_3), (x_2, x_4), (x_2, x_5), (x_3, x_1), (x_3, x_3), (x_3, x_4), (x_3, x_5), (x_4, x_2), (x_4, x_3),$

$$(x_5, x_1), (x_5, x_3)\}, R_2 = \{(x_1, x_2), (x_1, x_4), (x_1, x_5), (x_2, x_2), (x_2, x_3), (x_3, x_3), (x_4, x_3), (x_4, x_5)\}. \text{ Then } M_{R_1} = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{pmatrix},$$

$$M_{R_2} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, M_R = M_{R_1} \wedge M_{R_2} = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Definition 2.1 [16,37] Suppose U is a finite universe and A is a family of binary relations on U , then (U, A) is called a relation system. $(U, A = C \cup D)$ satisfying $C \cap D \neq \emptyset$ is referred to as a relation decision system, while C, D are called the conditional relation set and the decisional relation set of (U, A) respectively.

Suppose $E = \{R_1, R_2, \dots, R_m\}$ is a family of binary relations on U , the following operations are defined:

- (1) For any $R_i, R_j \in E$, $R_i \cap R_j = \{(x, y) \in U \times U : (x, y) \in R_i \text{ and } (x, y) \in R_j\}$;
- (2) $R_E = \bigcap_{R \in E} R = R_1 \cap R_2 \cdots \cap R_m = \{(x, y) \in U \times U : \forall R \in E, (x, y) \in R\}$.

Based on Definition 2.1, if $R_C = \bigcap_{R \in C} R \subseteq R_D = \bigcap_{R \in D} R$, then $(U, C \cup D)$ is referred to as a consistent relation decision system; otherwise, it is called an inconsistent relation decision system.

To characterize the consistency in a relation decision system, the notion of positive region given in Definition 2.2 is needed.

Definition 2.2 [16,37] Suppose $(U, C \cup D)$ is a relation decision system. Then $Pos_U(C, D) = \{x \in U : r_{R_C}(x) \subseteq r_{R_D}(x)\}$ is referred to as the C-positive region of D .

If $Pos_U(C, D) = \emptyset$, then $(U, C \cup D)$ is called totally inconsistent.

Note that the positive region defined in Definition 2.2 is a generalization of that given in [1,12], which can be used for the attribute reduction of decision systems with missing decision values. The following example shows the calculation process of the positive region in a relation decision system.

Example 2.2 Suppose $(U, C \cup D)$ is a relation decision system, where $U = \{x_1, x_2, x_3, x_4, x_5\}$, $C = \{R_1, R_2, R_3\}$ and $D = \{R_d\}$. The relation matrices corresponding to R_1, R_2, R_3 and R_d are as follows:

$$M_{R_1} = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{pmatrix}, M_{R_2} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{pmatrix}, M_{R_3} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \end{pmatrix}, M_{R_d} = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{pmatrix}. \text{ Based on}$$

U	a_1	a_2	a_3	a_4	d
x_1	1	2	*	1	1
x_2	0	1	1	0	1
x_3	0	1	*	0	2
x_4	1	2	1	0	1
x_5	*	0	0	0	1

Fig. 1. An incomplete decision system.

U	a_1	a_2	a_3	a_4	d
x_1	8.925	80.043	*	73.459	2
x_2	89.342	54.875	61.341	89.519	1
x_3	35.012	16.129	*	40.112	2
x_4	45.628	42.637	19.603	*	1
x_5	*	30.126	53.689	*	1

Fig. 2. An incomplete numerical decision system.

Definition 2.1, $\mathbf{M}_{R_C} = \bigwedge_{i=1}^3 \mathbf{M}_{R_i} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{pmatrix}.$

Then we have $r_{R_C}(x_1) = \{x_3\}$, $r_{R_C}(x_2) = \{x_3, x_4, x_5\}$, $r_{R_C}(x_3) = \{x_1, x_3\}$, $r_{R_C}(x_4) = \{x_2\}$, $r_{R_C}(x_5) = \{x_1, x_3\}$, $r_{R_D}(x_1) = \{x_3, x_4\}$, $r_{R_D}(x_2) = \{x_1, x_3, x_5\}$, $r_{R_D}(x_3) = \{x_1, x_3, x_5\}$, $r_{R_D}(x_4) = \{x_3\}$, $r_{R_D}(x_5) = \{x_1, x_3\}$. Therefore, $Pos_U(C, D) = \{x_1, x_3, x_5\}$.

Based on the definition of positive region, the following proposition holds.

Proposition 2.1 [16,37] Suppose $(U, C \cup D)$ is a relation decision system. Then.

(1) $(U, C \cup D)$ is a consistent relation decision system if and only if $Pos_U(C, D) = U$.

(2) If $B \subseteq C$, then $Pos_U(B, D) \subseteq Pos_U(C, D)$ holds.

Note that if D is a family of equivalence relations on U , then $Pos_U(C, D)$ degenerates into the positive region proposed by Wang [37]. Simultaneously, if C is a family of equivalence relations on U , then $Pos_U(C, D)$ degenerates into the positive region proposed by Pawlak [25,26]. Therefore, a relation decision system is a generalization of a decision system.

Definition 2.3 [16,37] Suppose $(U, C \cup D)$ is a relation decision system and $B \subseteq C$. B is called a reduction of $(U, C \cup D)$ if B satisfies the following conditions:

(1) $Pos_U(B, D) = Pos_U(C, D)$.

(2) For any $B' \subset B$, $Pos_U(B', D) \neq Pos_U(C, D)$.

Example 2.3 Following Example 2.2, let $B = \{R_1, R_2\}$. After calculation, we get $\mathbf{M}_{R_C} = \mathbf{M}_{R_B} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{pmatrix}$ and $\mathbf{M}_{R_D} = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{pmatrix}.$

It is obvious that $Pos_U(B, D) = Pos_U(C, D)$. In addition, it is verified that $Pos_U(\{R_1\}, D) \neq Pos_U(C, D)$ and $Pos_U(\{R_2\}, D) \neq Pos_U(C, D)$. Therefore, $B = \{R_1, R_2\}$ is a reduction of $(U, C \cup D)$.

Definition 2.4 [16,37] Suppose $(U, C \cup D)$ is a relation decision system and $B \subseteq C$. For any $R \in B$, if $Pos_U(B, D) = Pos_U(B - \{R\}, D)$, then R is referred to as a redundant relative to D in B . Otherwise R is indispensable.

Definition 2.4 shows that removing a redundant conditional relation will not change the original positive region. This plays a crucial role in a reduction algorithm. An example is given to illustrate Definition 2.4 as follows.

Example 2.4 Following Example 2.3, since $B = \{R_1, R_2\}$ and $\mathbf{M}_{R_C} = \mathbf{M}_{R_B}$. According to Definition 2.2, we have $Pos_U(C, D) = Pos_U(C - \{R_3\}, D) = \{x_1, x_3, x_5\}$. Hence R_3 is referred to as a redundant relative to D in C .

Definition 2.5 [16,37] Suppose $(U, C \cup D)$ is a relation decision system and $B \subseteq C$. For any $R \in B$, the significance of R relative to D is defined by $Sig(R, B, D) = |Pos_U(B, D) - Pos_U(B - \{R\}, D)|$.

Definition 2.5 is employed in the process of searching for a reduction. Obviously, for a conditional relation R , if $Sig(R, B, D) = 0$, then the conditional relation R is redundant with respect to the condition subset B ; otherwise it is essential.

U	a_1	a_2	a_3	a_4	d
x_1	$\{1, 2\}$	$\{0, 1\}$	$\{1\}$	$\{1\}$	1
x_1	$\{2\}$	$\{1\}$	$\{2\}$	$\{1\}$	2
x_1	$\{0, 2\}$	$\{0\}$	$\{0, 1, 2\}$	$\{1, 2\}$	2
x_1	$\{2\}$	$\{0, 1\}$	$\{1, 2\}$	$\{2\}$	0
x_1	$\{0\}$	$\{0\}$	$\{0\}$	$\{1\}$	1

Fig. 3. An incomplete set-valued decision system.

U	a_1	a_2	a_3	a_4	d
x_1	1	2	2	1	1
x_2	0	1	1	0	2
x_3	0	2	2	1	2
x_4	1	2	1	0	1
x_5	1	0	0	0	1

Fig. 4. A decision information system.

3. Updating positive region when conditional relation set has varied

The algorithm given in [16] has greater uniformity and generality than the other existing algorithms. In fact, it is a reduction method based on the positive region of a relation decision systems, and can be applied to attribute reduction of all kinds of decision systems with missing decision value, such as complete or incomplete decision systems, set-valued decision systems, and covering decision systems. Obviously, it is more general to various decision systems. However, for a relation decision system with dynamic conditional relation set, it is time-consuming. In this section, we discuss how to employ the original positive region to update the new positive region when the conditional relation set of a relation decision system has varied.

The changes of the conditional relation set of a relation decision system include two cases: increasing and decreasing. We will conduct our discussion with respect to these two cases.

Firstly, the definition of the characteristic vector of a set is presented, which is employed in the calculation of the positive region later.

Definition 3.1 [19] Given $U = \{x_1, x_2, \dots, x_n\}$ and $X \subseteq U$. The characteristic vector $\lambda(X)$ of X is defined as follows:

$\lambda(X) = (\lambda_1, \lambda_2, \dots, \lambda_n)^T$, where $\lambda_i = \begin{cases} 1, & x_i \in X \\ 0, & x_i \notin X \end{cases}$, T is the transpose operation of matrices.

Moreover, $\lambda(X^c) = (u_1, u_2, \dots, u_n)^T$, $u_i = \begin{cases} 1, & \lambda_i = 0 \\ 0, & \lambda_i = 1 \end{cases}$, where X^c is the complement of X .

In the following, we give an example to illustrate the above concepts.

Example 3.1 Given $U = \{x_1, x_2, x_3, x_4, x_5, x_6\}$, and $X = \{x_1, x_2, x_4\}$. According to Definition 3.1, it can be obtained that $\lambda(X) = (1, 1, 0, 1, 0, 0)^T$ and $\lambda(X^c) = (0, 0, 1, 0, 1, 1)^T$.

The following proposition states that the operations between characteristic vectors can be employed to express the inclusion relationships between sets.

Proposition 3.1 Given $U = \{x_1, x_2, \dots, x_n\}$ and $X, Y \subseteq U$. $X \subseteq Y$ if and only if $\lambda^T(X) \circ \lambda(Y^c) = 0$, where $\circ$ stands for the product operation of characteristic vectors.

Proof. Sufficiency. Let $X \subseteq Y$. The i -th components of $\lambda^T(X)$ and $\lambda(Y)$ are denoted as $\lambda^T(X)(i)$ and $\lambda(Y)(i)$ ($1 \leq i \leq n$) respectively. Then we have that if $\lambda^T(X)(i) = 1$, then $\lambda(Y)(i) = 1$ and $\lambda(Y^c)(i) = 0$, hence $\lambda^T(X) \circ \lambda(Y^c) = 0$.

Necessity. Let $\lambda^T(X) \circ \lambda(Y^c) = 0$. For any $x_i \in X$ ($1 \leq i \leq n$), we have $\lambda^T(X)(i) = 1$. Since $\lambda^T(X) \circ \lambda(Y^c) = 0$, we get $\lambda(Y^c)(i) = 0$, which implies that $\lambda(Y)(i) = 1$, namely $x_i \in Y$. Therefore, $X \subseteq Y$.

Combined with Proposition 3.1, through vector operation, we get the following theorem which gives another characterization of the positive region.

Theorem 3.1 Suppose $(U, C \cup D)$ is a relation decision system and $B \subseteq C$. Then

$$Pos_U(B, D) = \{x \in U : \lambda^T(r_{R_B}(x)) \circ \lambda((r_{R_D}(x))^c) = 0\}.$$

Proof. According to Definition 2.2 and Proposition 3.1, the result is clearly established.

3.1. Updating positive region when conditional relation set has increased

When the conditional relation set has increased in a relation decision system, how to incrementally update the positive region based on the previous positive region is the task of this subsection. In this subsection, the incremental updating mechanism of the positive region is designed for a relation decision system with an increased conditional relation set. Since the calculation of the positive region is closely related to the right neighborhood in a relation decision system, we first give an incremental updating method for the right neighborhood of an object when the conditional relation set has increased.

Proposition 3.2 Suppose $(U, C \cup D)$ is a relation decision system, and $x \in U$, $B, Q \subseteq C, B \cap Q = \emptyset$. Then $r_{R_{B \cup Q}}(x) = r_{R_B}(x) - \{y \in r_{R_B}(x) : \exists R \in Qs.t. y \notin r_R(x)\}$.

Proof. According to Theorem 2.1, $r_{R_{B \cup Q}}(x) = r_{R_B}(x) \cap r_{R_Q}(x) = \bigcap_{Q \in B \cup Q} R_Q(x)$. It is obvious that $r_{R_{B \cup Q}}(x) \subseteq r_{R_B}(x)$. In fact, we need only prove that $r_{R_B}(x) = r_{R_{B \cup Q}}(x) \cup \{y \in r_{R_B}(x) : \exists R \in Qs.t. y \notin r_R(x)\}$.

On one hand, for any $z \in r_{R_B}(x)$, since $r_{R_{B \cup Q}}(x) \subseteq r_{R_B}(x)$, two cases need to be considered:

Case 1: If $z \in r_{R_{B \cup Q}}(x)$, then it is obvious that $r_{R_{B \cup Q}}(x) \cup \{y \in r_{R_B}(x) : \exists R \in Qs.t.y \notin r_R(x)\}$.

Case 2: If $z \in r_{R_B}(x) - r_{R_{B \cup Q}}(x)$, then $z \in r_{R_B}(x)$, but $z \notin r_{R_{B \cup Q}}(x) = \bigcap_{a \in B \cup Q} R_a(x)$, namely $z \in \{y \in r_{R_B}(x) : \exists R \in Qs.t.y \notin r_R(x)\}$.

Hence, we have $r_{R_B}(x) \subseteq r_{R_{B \cup Q}}(x) \cup \{y \in r_{R_B}(x) : \exists R \in Qs.t.y \notin r_R(x)\}$.

On the other hand, for any $z \in r_{R_{B \cup Q}}(x) \cup \{y \in r_{R_B}(x) : \exists R \in Qs.t.y \notin r_R(x)\}$, since $r_{R_{B \cup Q}}(x) \subseteq r_{R_B}(x)$, it is obvious that $z \in r_{R_B}(x)$.

To sum up, $r_{R_{B \cup Q}}(x) = r_{R_B}(x) - \{y \in r_{R_B}(x) : \exists R \in Qs.t.y \notin r_R(x)\}$.

Based on the result of Proposition 3.2, we obtain the following theorem where an incremental updating method for the positive region is given when some conditional relations are added to a relation decision system. Compared with the non-incremental method of computing the new positive region, this new positive region calculation method can avoid some repeated calculations which will be verified in the experimental study.

Theorem 3.2 Suppose $(U, C \cup D)$ is a relation decision system, and $x \in U, B, Q \subseteq C, B \cap Q = \emptyset$. Then

$$Pos_U(B \cup Q, D) = Pos_U(B, D) \cup \{y \in U - Pos_U(B, D) : \lambda^T(r_{R_{B \cup Q}}(x))^\circ \lambda((r_{R_D}(x))^C) = 0\}$$

Proof. Let $N = \{y \in U - Pos_U(B, D) : \lambda^T(r_{R_{B \cup Q}}(x))^\circ \lambda((r_{R_D}(x))^C) = 0\}$. According to Definition 2.2, we have $Pos_U(B \cup Q, D) = \{x \in U : r_{R_{B \cup Q}}(x) \subseteq r_{R_D}(x)\}$.

On one hand, we first prove that $Pos_U(B, D) \cup N \subseteq Pos_U(B \cup Q, D)$. Then two cases need to be considered:

Case 1: For any $x \in Pos_U(B, D)$, according to (2) of Proposition 2.1, $Pos_U(B, D) \subseteq Pos_U(B \cup Q, D)$, so $x \in Pos_U(B \cup Q, D)$.

Case 2: For any $x \in N$, we have $\lambda^T(r_{R_{B \cup Q}}(x))^\circ \lambda((r_{R_D}(x))^C) = 0$, namely $r_{R_{B \cup Q}}(x) \subseteq r_{R_D}(x)$, which implies that $x \in Pos_U(B \cup Q, D)$.

To sum up, $Pos_U(B, D) \cup N \subseteq Pos_U(B \cup Q, D)$.

On the other hand, for any $x \in Pos_U(B \cup Q, D)$, according to (2) of Proposition 2.1, we have $Pos_U(B, D) \subseteq Pos_U(B \cup Q, D)$. If $x \in Pos_U(B, D)$, then $x \in Pos_U(B, D) \cup N$. If $x \in Pos_U(B \cup Q, D) - Pos_U(B, D)$, then we have $r_{R_B}(x) \not\subseteq r_{R_D}(x)$ but $r_{R_{B \cup Q}}(x) \subseteq r_{R_D}(x)$, namely $\lambda^T(r_{R_{B \cup Q}}(x))^\circ \lambda((r_{R_D}(x))^C) = 0$. Hence, it can be got $x \in Pos_U(B, D) \cup \{y \in U - Pos_U(B, D) : \lambda^T(r_{R_{B \cup Q}}(x))^\circ \lambda((r_{R_D}(x))^C) = 0\}$.

So $Pos_U(B \cup Q, D) \subseteq Pos_U(B, D) \cup \{y \in U - Pos_U(B, D) : \lambda^T(r_{R_{B \cup Q}}(x))^\circ \lambda((r_{R_D}(x))^C) = 0\}$.

In particular, when the adding conditional relation set contains only one relation, the following result can be obtained.

Corollary 3.1 Suppose $(U, C \cup D)$ is a relation decision system, and $x \in U, B \subseteq C, R \in C, R \notin B$. Then

$$Pos_U(B \cup \{R\}, D) = Pos_U(B, D) \cup \{y \in U - Pos_U(B, D) : \lambda^T(r_{R_{B \cup \{R\}}}(x))^\circ \lambda((r_{R_D}(x))^C) = 0\}$$

$$r_{R_{B \cup \{R\}}}(x) = r_{R_B}(x) - \{y \in r_{R_B}(x) : y \notin r_R(x)\}$$

In order to facilitate the description of the proposed updating method of the positive region, we see the following example.

Example 3.2 Suppose $(U, C \cup D)$ is a relation decision system. $U = \{x_1, x_2, x_3, x_4, x_5\}$, $C = \{R_1, R_2, R_3, R_4\}$ and $D = \{R_d\}$. Let $B = \{R_1, R_3\}$, $Q = \{R_2, R_4\}$. The relation matrices corresponding to relation R_1, R_2, R_3, R_4 and R_D are as follows:

$$\begin{aligned} M_{R_1} &= \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{pmatrix}, M_{R_2} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{pmatrix}, M_{R_3} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix}, \\ M_{R_4} &= \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}, M_{R_D} = \begin{pmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{pmatrix}. \text{ We compute } M_{R_B} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{pmatrix} \end{aligned}$$

When the conditional relation set B is changed to $B \cup Q$, the new positive region is calculated in the original positive region. First calculate $r_{R_B}(x_1) = \{x_1, x_3\}$, $r_{R_B}(x_2) = \{x_3, x_4\}$, $r_{R_B}(x_3) = \{x_1, x_3, x_5\}$, $r_{R_B}(x_4) = \{x_2\}$, $r_{R_B}(x_5) = \{x_1, x_2, x_3\}$, $r_{R_D}(x_1) = \{x_3, x_4\}$, $r_{R_D}(x_2) = \{x_1, x_3, x_5\}$, $r_{R_D}(x_3) = \{x_1, x_3, x_5\}$, $r_{R_D}(x_4) = \{x_2, x_3\}$, $r_{R_D}(x_5) = \{x_1, x_3\}$ and $Pos_U(B, D) = \{x_3, x_4\}$. According to Proposition 3.2 and Theorem 3.2, we have $r_{R_{B \cup Q}}(x_1) = r_{R_B}(x_1) - \{x_1\} = \{x_3\}$, $r_{R_{B \cup Q}}(x_2) = r_{R_B}(x_2) - \{x_3\} = \{x_4\}$, $r_{R_{B \cup Q}}(x_3) = r_{R_B}(x_3) - \emptyset = \{x_1, x_3, x_5\}$, $r_{R_{B \cup Q}}(x_4) = r_{R_B}(x_4) - \emptyset = \{x_2\}$, $r_{R_{B \cup Q}}(x_5) = r_{R_B}(x_5) - \{x_1, x_2\} = \{x_3\}$ and $Pos_U(B \cup Q, D) = \{x_1, x_3, x_4, x_5\}$.

Example 3.2 shows that when the conditional relation set has increased, the right neighborhood and positive region can be updated based on the original calculation results according to Proposition 3.2 and Theorem 3.2, which can avoid some repeated calculations.

3.2. Updating positive region when conditional relation set has decreased

In this subsection, we discuss the updating method of the new positive region in the relation decision system whose conditional relation set has decreased. Similarly, we first put forward the update mechanism of the new right neighbor when some conditional relations are removed from the relation decision system.

Proposition 3.3 Suppose $(U, C \cup D)$ is a relation decision system, and $x \in U$, $Q \subseteq B \subseteq C$. Then $r_{R_B-Q}(x) = r_{R_B}(x) \cup \{y \in U - r_{R_B}(x) : \forall R \in B - Q, s.t. y \in r_R(x)\}$.

Proof. On one hand, for any $z \in r_{R_B-Q}(x)$, since $r_{R_B}(z) \subseteq r_{R_B-Q}(z)$, then two cases need to be considered: Case 1: If $z \in r_{R_B}(x)$, then $z \in r_{R_B}(x) \cup \{y \in U - r_{R_B}(x) : \forall R \in B - Q, s.t. y \in r_R(x)\}$.

Case 2: If $z \notin r_{R_B}(x)$ and $z \in r_{R_B-Q}(x)$, namely $z \in \bigcap_{a \in B-Q} R_a(x)$, which implies that $z \in \{y \in U - r_{R_B}(x) : \forall R \in B - Q, s.t. y \in r_R(x)\}$.

Hence $r_{R_B-Q}(x) \subseteq r_{R_B}(x) \cup \{y \in U - r_{R_B}(x) : \forall R \in B - Q, s.t. y \in r_R(x)\}$.

On the other hand, for any $z \in r_{R_B}(x) \cup \{y \in U - r_{R_B}(x) : \forall R \in B - Q, s.t. y \in r_R(x)\}$, it is obvious that $z \in r_{R_B-Q}(x)$, which implies that $r_{R_B}(x) \cup \{y \in U - r_{R_B}(x) : \forall R \in B - Q, s.t. y \in r_R(x)\} \subseteq r_{R_B-Q}(x)$.

To sum up, $r_{R_B-Q}(x) = r_{R_B}(x) \cup \{y \in U - r_{R_B}(x) : \forall R \in B - Q, s.t. y \in r_R(x)\}$.

Through the updating mechanism of the right neighborhood given in Proposition 3.3, the following theorem gives the incremental calculation method of the new positive region. The experiments to be carried out in the experimental studies show that the calculation time based on this incremental method is significantly lower than that of the non-incremental method.

Theorem 3.3 Suppose $(U, C \cup D)$ is a relation decision system, and $x \in U$, $B, Q \subseteq C, B \cap Q = \emptyset$. Then $Pos_U(B - Q, D) = Pos_U(B, D) - \{y \in Pos_U(B, D) : \lambda^T(r_{R_B-Q}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0\}$.

Proof. Since $\lambda^T(r_{R_B-Q}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0$, then $r_{R_B-Q}(x) \not\subseteq r_{R_D}(x)$, which implies $x \notin Pos_U(B - Q, D)$. In the fact, we need only to prove $Pos_U(B, D) = Pos_U(B - Q, D) \cup \{y \in Pos_U(B, D) : \lambda^T(r_{R_B-Q}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0\}$.

On one hand, for any $x \in Pos_U(B, D)$, according to (2) of Proposition 2.1, we have $Pos_U(B - Q, D) \subseteq Pos_U(B, D)$. Two cases should be considered:

Case 1: If $x \in Pos_U(B - Q, D)$, then we have $x \in Pos_U(B - Q, D) \cup \{y \in Pos_U(B, D) : \lambda^T(r_{R_B-Q}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0\}$.

Case 2: If $x \notin Pos_U(B - Q, D)$ but $x \in Pos_U(B, D)$, then we have $x \in \{y \in Pos_U(B, D) : \lambda^T(r_{R_B-Q}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0\}$. Hence, $Pos_U(B, D) \subseteq Pos_U(B - Q, D) \cup \{y \in Pos_U(B, D) : \lambda^T(r_{R_B-Q}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0\}$.

On the other hand, for any $x \in Pos_U(B - Q, D) \cup \{y \in Pos_U(B, D) : \lambda^T(r_{R_B-Q}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0\}$, because $Pos_U(B - Q, D) \subseteq Pos_U(B, D)$, it obvious that $x \in Pos_U(B, D)$, which implies that $Pos_U(B - Q, D) \cup \{y \in Pos_U(B, D) : \lambda^T(r_{R_B-Q}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0\} \subseteq Pos_U(B, D)$.

To sum up, we have $Pos_U(B - Q, D) = Pos_U(B, D) - \{y \in Pos_U(B, D) : \lambda^T(r_{R_B-Q}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0\}$.

When there is only one conditional relation to be removed in a relation decision system, the following corollary holds.

Corollary 3.2 Suppose $(U, C \cup D)$ is a relation decision system, and $x \in U$, $R \in B \subseteq C$. Then $Pos_U(B - \{R\}, D) = Pos_U(B, D) - \{y \in Pos_U(B, D) : \lambda^T(r_{R_B-\{R\}}(x))^\circ \lambda((r_{R_D}(x))^C) \neq 0\}$, where $r_{R_B-\{R\}}(x) = r_{R_B}(x) \cup \{y \in U - r_{R_B}(x) : \forall R \in B - \{R\}, s.t. y \in r_R(x)\}$.

For the convenience of explanation, we take the following example for showing the update mechanism from the original positive region to the new positive region in the case of the conditional relation set decreases.

Example 3.3 Following Example 3.2, let $B = \{R_1, R_2, R_3, R_4\}$, $Q = \{R_2, R_4\}$. Then we have $r_{R_B}(x_1) = \{x_3\}$, $r_{R_B}(x_2) = \{x_4\}$, $r_{R_B}(x_3) = \{x_1, x_3, x_5\}$, $r_{R_B}(x_4) = \{x_2\}$, $r_{R_B}(x_5) = \{x_3\}$, and $Pos_U(B, D) = \{x_1, x_3, x_4, x_5\}$. According to Proposition 3.3, we have $r_{R_B-Q}(x_1) = r_{R_B}(x_1) \cup \{x_1\} = \{x_1, x_3\}$, $r_{R_B-Q}(x_2) = r_{R_B}(x_2) \cup \{x_3\} = \{x_3, x_4\}$, $r_{R_B-Q}(x_3) = r_{R_B}(x_3) \cup \emptyset = \{x_1, x_3, x_5\}$, $r_{R_B-Q}(x_4) = r_{R_B}(x_2) \cup \emptyset = \{x_2\}$, $r_{R_B-Q}(x_5) = r_{R_B}(x_5) \cup \{x_1, x_2\} = \{x_1, x_2, x_3\}$. According to Theorem 3.3, $Pos_U(B - Q, D) = Pos_U(B, D) - \{x_1, x_5\} = \{x_3, x_4\}$.

The above example presents that when the conditional relation set has decreased, the right neighborhood and positive region can be updated based on the original result of calculation according to Proposition 3.3 and Theorem 3.3, and some unnecessary calculations can be avoided.

4. The non-incremental and incremental updating reduction algorithms for dynamic relation decision systems

Based on the positive region incremental updating mechanism given in Section 3, the non-incremental and incremental updating reduction algorithms for a relation decision system with a dynamic conditional relation set are developed in this section.

Firstly, we design a non-incremental updating reduction algorithm for relation decision system whose conditional relation sets have varied, named Algorithm 1.

Algorithm 1 A non-incremental updating reduction algorithm for a relation decision system with dynamic conditional relation set

Input: A relation decision system $(U, C \cup D)$, the adding conditional relation set C_{in} (or the removing conditional relation set C_{de}), where $C_{in} \cap C = \emptyset$ and $\emptyset \neq C_{de} \subset C$.

Output: A new reduction Red of $(U, \tilde{C} \cup D)$, where $\tilde{C} = C \cup C_{in}$ (or $\tilde{C} = C - C_{de}$).

Begin

1. Initialize $\tilde{C} \leftarrow C \cup C_{in}$ (or $C - C_{de}$);
2. Compute $Pos_U(\tilde{C}, D)$; // According to Definition 2.2 and Theorem 3.1
3. Compute $Sig(R_i, \tilde{C}, D)$ for each conditional relation $R_i \in \tilde{C}$; // According to Definition 2.5
4. Use the results in step 3 to construct an ascending sequence and denote it as $\{\tilde{R}_1, \tilde{R}_2, \dots, \tilde{R}_{|\tilde{C}|}\}$;
5. Let $Red \leftarrow \tilde{C}$, $N \leftarrow |\tilde{C}|$;
6. **For** $i = 1$ to N **do**
7. Compute $Pos_U(\tilde{C} - \{\tilde{R}_i\}, D)$; // According to Definition 2.2 and Theorem 3.1
8. **If** $Pos_U(\tilde{C} - \{\tilde{R}_i\}, D) = Pos_U(\tilde{C}, D)$ **then** // According to Definition 2.4
9. $Red \leftarrow Red - \{\tilde{R}_i\}$ and $\tilde{C} \leftarrow \tilde{C} - \{\tilde{R}_i\}$;
10. **End if**
11. **End for**
12. **Return** Red .

End

Let $(U, C \cup D)$ be a relation decision system. The adding conditional relation set is denoted as C_{in} while the removing conditional relation set is denoted as C_{de} . Suppose $\tilde{C} = C \cup C_{in}$ (or $\tilde{C} = C - C_{de}$), then $|\tilde{C}| = |C| + |C_{in}|$ (or $|\tilde{C}| = |C| - |C_{de}|$), where $|Y|$ denotes the cardinality of Y .

The calculation in Algorithm 1 includes the following parts:

- (1) Calculate the new positive region according to Theorem 3.1 when add or remove some conditional relations in the relation decision system, whose time complexity is $O(|\tilde{C}|^2 |U|^2)$.
- (2) Calculate the significance of each conditional relation to the new conditional relation set, and sorts them in ascending order in terms of Definition 2.5, whose time complexity is $O(|\tilde{C}|^2 |U|^2)$. In the sorting, if the significances of two or more conditional relations is equal, they are sorted according to the subscript size of their corresponding conditional relations.
- (3) Calculate a reduction of the relation decision system $(U, C \cup D)$ by removing some redundant conditional relations from $(U, C \cup D)$ according to Definition 2.4, whose time complexity is $O(|\tilde{C}|^2 |U|^2)$.

Hence the time complexity of Algorithm 1 is $O(|\tilde{C}|^2 |U|^2)$.

Note that most reduction methods are forward search and greedy algorithms [11,13,15,29,49] that initialize empty sets as initialization and iteratively increase conditional relation with great significance until the positive region no longer varies, this process is time-consuming. If these methods of reduction are used to compute the reduction of a relation decision system, the time complexity of this process is $O(|\tilde{C}|^3 |U|^3)$, which is more than that of Algorithm 1. Therefore, Algorithm 1 performs better than the forward search or greedy search algorithms in computing a reduction in a relation decision system.

For a relation decision system with a dynamic conditional relation set, Algorithm 1 updates the reduction of the relation decision system by some recalculation, which is time-consuming.

Algorithm 2 is an incremental updating reduction algorithm when C_{in} is added into the original relation decision system. According to Theorems 3.2, based on the original positive region and the updating mechanism for new right neighborhood, the new positive region is updated incrementally. It is worth noting that if the new positive region is equal to the original positive region, then the original reduction is directly output as the reduction of the new relation decision system.

The calculation in Algorithm 2 includes the following parts:

Algorithm 2 An incremental updating reduction algorithm for a relation decision system whose conditional relation set has increased

Input: A relation decision system $(U, C \cup D)$, the former reduction Red , the former $Pos_U(C, D)$ and the adding conditional relation set C_{in} , where $C_{in} \cap C = \emptyset$.

Output: A new reduction $\tilde{Red}$ of $(U, \tilde{C} \cup D)$, where $\tilde{C} = C \cup C_{in}$.

Begin

1. Initialize $\tilde{C} \leftarrow C \cup C_{in}$;
2. Compute $Pos_U(\tilde{C}, D)$; // According to Theorem 3.2
3. **If** $Pos_U(\tilde{C}, D) = Pos_U(C, D)$, **then**
4. $Red \leftarrow Red$, turn to step 15;
5. **End if**
6. Compute $Sig(R_i, \tilde{C}, D)$ for each conditional relation $R_i \in \tilde{C}$; // According to Definition 2.5
7. Use the results in step 6 to construct an ascending sequence and denote it as $\{\tilde{R}_1, \tilde{R}_2, \dots, \tilde{R}_{|\tilde{C}|}\}$;
8. Let $Red \leftarrow C$, $N \leftarrow |\tilde{C}|$;
9. **For** $i = 1$ to N **do**
10. Compute $Pos_U(\tilde{C} - \{\tilde{R}_i\}, D)$; // According to Theorem 3.2
11. **If** $Pos_U(\tilde{C} - \{\tilde{R}_i\}, D) = Pos_U(C, D)$, **then** // According to Definition 2.4
12. $Red \leftarrow Red - \{\tilde{R}_i\}$ and $\tilde{C} \leftarrow \tilde{C} - \{\tilde{R}_i\}$;
13. **End if**
14. **End for**
15. **Return** $\tilde{Red}$.

End

- (1) Calculate the new positive region according to Theorem 3.2, which is the key step of Algorithm 2, and its time complexity is $O(|C_{in}||U|^2)$.
- (2) Judge whether the new positive region is equal to the original positive region. If they are equal, the original reduction is output as the new reduction, otherwise the original reduction needs to be updated. The time complexity of this process is $O(|C \cup C_{in}|)$.
- (3) Calculate the significance of each conditional relation in the new conditional relation set, and sorts them in ascending order (The sorting method is similar to Algorithm 1), whose time complexity is not more than $O(|C \cup C_{in}||C_{in}||U|^2)$.
- (4) Calculate a reduction of the relation decision system when the conditional relation set increases by removing some redundant conditional relations according to Definition 2.4. The worst time complexity of this process is $O(|C \cup C_{in}||C_{in}||U|^2)$.

Hence the total time complexity of Algorithm 2 is not more than $O(|C \cup C_{in}||C_{in}||U|^2)$.

Algorithm 3 is an incremental updating reduction algorithm when C_{de} is removed from the original relation decision system.

Algorithm 3 An incremental updating reduction algorithm for a relation decision system whose conditional relation set has decreased

Input: A relation decision system $(U, C \cup D)$, the former reduction Red , the former $Pos_U(C, D)$ and the removing conditional relation set C_{de} , where $\emptyset \neq C_{de} \subset C$.

Output: A new reduction $\tilde{Red}$ of $(U, \tilde{C} \cup D)$, where $\tilde{C} = C - C_{de}$.

Begin

1. Initialize $\tilde{C} \leftarrow C - C_{de}$;
2. Compute $Pos_U(\tilde{C}, D)$; // According to Theorem 3.3
3. **If** $C_{de} \cap Red = \emptyset$, **then**
4. $Red \leftarrow Red$, turn to step 15;
5. **End if**
6. Compute $Sig(R_i, \tilde{C}, D)$ for each conditional relation $R_i \in \tilde{C}$; // According to Definition 2.5
7. Use the results in step 6 to construct an ascending sequence and denote it as $\{\tilde{R}_1, \tilde{R}_2, \dots, \tilde{R}_{|\tilde{C}|}\}$;
8. Let $Red \leftarrow C$, $N \leftarrow |\tilde{C}|$;

(continued on next page)

a (continued)

Algorithm 3 An incremental updating reduction algorithm for a relation decision system whose conditional relation set has deceased

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9. For  $i = 1$  to  $N$  do
10. Compute  $Pos_U(\tilde{C} - \{R_i\}, D)$ ; // According to Theorem 3.3
11. If  $Pos_U(\tilde{C} - \{R_i\}, D) = Pos_U(\tilde{C}, D)$  then // According to Definition 2.4
12.  $Red \leftarrow Red - \{R_i\}$  and  $\tilde{C} \leftarrow \tilde{C} - \{R_i\}$ ;
13. End if
14. End for
15. Return  $Red$ .
End

```

According to Theorem 3.3, an incremental updating of the original positive region is proposed. In this process, if the removing conditional relation set does not intersect with the original reduction, then the original reduction is output as a new reduction, otherwise the original reduction need be updated incrementally.

The calculation in Algorithm 3 includes the following parts:

- (1) Calculate the new positive region according to Theorem 3.3, which is the key step of Algorithm 3, and its time complexity is $O(|C - C_{de}||U|^2)$.
- (2) Judge whether the new positive region is equal to the original positive region when some conditional relations are decreased. If they are equal, the original reduction is output as the new reduction, otherwise the original reduction needs to be updated. The time complexity of this process is $O(|C - C_{de}|)$.
- (3) Calculate the significance of each conditional relation in the new conditional relation set and sorts them in ascending order (The sorting method is similar to Algorithm 1). The time complexity of this process is $O(|C - C_{de}||U|^2)$.
- (4) Calculate a reduction of the relation decision system when the conditional relation set decreases by removing some redundant conditional relations according to Definition 2.4. The time complexity of this process is not more than $O(|C - C_{de}||U|^2 + |C - C_{de}||U|)$.

Hence the time complexity of Algorithm 3 is not more than $O(|C - C_{de}||U|^2 + |C - C_{de}||U|)$.

In reality, the number of objects in decision systems is often larger than the dimension in most cases, especially for a large-scale decision system. Moreover, the adding conditional relation set C_{in} and the removing conditional relation set C_{de} are much smaller than the original conditional relation set. In view of the above analysis, Algorithm 2 and Algorithm 3 are more efficient in computing reductions than Algorithm 1.

5. Experimental studies

In this section, to demonstrate the effectiveness of our proposed incremental updating algorithm, some comparative experiments are designed on nine datasets with missing values. The datasets are downloaded from UCI's Machine Learning Database Repository (<https://www.ics.uci.edu/mllearn/MLRepository.html>), which are listed in Table 1. All experiments are performed on computer with Intel Xeon Gold 6136 CPU 3.0GHZ, 256.0 GB of memory, running Win 10, and Algorithms are coded in MATLAB 2020.

Being able to produce the reduction of a dynamic decision systems with a nonequivalence decision relation is a distinct characteristic of our proposed incremental reduction algorithms. To illustrate this, the experiments are carried out on decision systems with missing decision values. To have such experimental datasets, we randomly changed some decision values in nine data sets into missing values. The concrete process of inducing relation decision system with a decision system with missing decision values is as follows.

Table 1

The description of data sets.

Data Sets	Samples	Attributes	Classes
Congressional voting records (CVR)	435	16	2
Soybean-large (Soybean)	307	35	19
Car Evaluation (Car)	1728	7	4
Dermatology	336	34	6
Tic-Tac-Toe Endgame (Tic-Tac-Toe)	958	10	2
Mushroom	8124	22	2
In-vehicle coupon recommendation (Recommendation)	12,684	23	4
Letter-recognition (Letter)	20,000	16	26
Chess-KR-VS-KK (Chess)	3199	37	2

Let $(U, A \cup \{d\})$ be a decision system with missing decision values, where $A = \{a_1, a_2, \dots, a_m\}$ is the conditional attribute set and d is the decisional attribute.

(1) If $a \in A$ is a symbolic attribute, then binary relations R_a and R_d on U can be induced via:

$$x_i R_a x_j \iff a(x_i) = a(x_j) \text{ or } a(x_i) = * \text{ or } a(x_j) = * \text{ and } x_i R_d x_j \iff d(x_i) = d(x_j) \text{ or } d(x_i) = * \text{ or } d(x_j) = * \text{ for any } x_i, x_j \in U;$$

(2) If $a \in A$ is numerical attribute, for a given threshold value $\partial > 0$, then binary relations R_a and R_d on U can be induced via:

$$x_i R_a x_j \iff |a(x_i) - a(x_j)| \leq \partial \text{ or } a(x_i) = * \text{ or } a(x_j) = * \text{ and } x_i R_d x_j \iff d(x_i) = d(x_j) \text{ or } d(x_i) = * \text{ or } d(x_j) = * \text{ for any } x_i, x_j \in U;$$

where $*$ denotes a missing attribute value and $a(x), d(x) (x \in U)$ denote the value of conditional attribute a and decisional attribute d of object x , respectively.

Let $C = \{R_{a_1}, R_{a_2}, \dots, R_{a_m}\}$ and $D = \{R_d\}$, then a relation decision system $(U, C \cup D)$ can be obtained in this way.

5.1. Comparison of non-incremental and incremental updating reduction algorithms when conditional relation set has increased

To show the performance of the proposed incremental updating reduction algorithm (Algorithm 2), experiments are carried out on nine datasets (in Table 1) according to non-incremental updating reduction algorithm (Algorithm 1) and Algorithm 2 when the conditional relation set has increased. To distinguish the computational efficiency of algorithms under different numbers of conditional relations, we first divide the conditional relation set of each dataset into 10 equal parts. Then add the first part to empty conditional relation set resulting in the first data set (Nos. 1), add the second part to the first data set resulting in the second data set (Nos. 2). In the same way, we obtain 10 data sets: Nos. 1, Nos. 2, ..., Nos. 10. It is obvious that the size of conditional relation set is gradually increased by 10 %.

Table 2 and Fig. 5 show the comparison of the computational time of Algorithm 1 and Algorithm 2 in each data set (Nos. 1–Nos. 10). In each sub-graph of Fig. 5, the x-coordinate represents the size of the conditional relation set, and the y-coordinate represents the computation time.

Table 2

Comparisons of computation times under non-incremental and incremental updating reduction algorithms over nine data sets when conditional relation set has increased.

Nos.	CVR			Soybean			Car		
	Alg-1	Alg-2	PDCT	Alg-1	Alg-2	PDCT	Alg-1	Alg-2	PDCT
1	0.049	0.050	−2.20 %	0.042	0.040	4.57 %	0.527	0.503	4.57 %
2	0.065	0.020	69.50 %	0.065	0.031	52.59 %	0.558	0.038	93.16 %
3	0.101	0.038	62.45 %	0.100	0.038	61.42 %	0.821	0.333	59.43 %
4	0.117	0.027	77.29 %	0.125	0.054	57.02 %	0.822	0.051	93.84 %
5	0.153	0.046	69.61 %	0.165	0.066	60.12 %	1.083	0.312	71.23 %
6	0.194	0.072	62.78 %	0.193	0.063	67.38 %	1.393	0.427	69.34 %
7	0.216	0.062	71.49 %	0.232	0.040	82.70 %	1.371	0.054	96.07 %
8	0.251	0.082	67.22 %	0.264	0.034	87.30 %	1.676	0.461	72.49 %
9	0.269	0.038	85.72 %	0.307	0.042	86.22 %	1.679	0.060	96.41 %
10	0.302	0.094	68.80 %	0.339	0.036	89.41 %	1.963	0.510	74.03 %
Nos.	Dermatology			Tic-Tac-Toe			Mushroom		
	Alg-1	Alg-2	PDCT	Alg-1	Alg-2	PDCT	Alg-1	Alg-2	PDCT
1	0.049	0.049	0.24 %	0.009	0.007	13.99 %	26.301	26.841	−2.05 %
2	0.098	0.064	34.82 %	0.015	0.007	51.69 %	45.187	22.598	49.99 %
3	0.134	0.061	54.41 %	0.019	0.009	51.58 %	74.754	33.547	55.12 %
4	0.193	0.080	58.60 %	0.024	0.009	60.34 %	93.791	25.903	72.38 %
5	0.233	0.083	64.22 %	0.033	0.011	66.20 %	114.739	17.063	85.13 %
6	0.278	0.093	66.48 %	0.033	0.005	84.15 %	135.807	17.632	87.02 %
7	0.334	0.063	81.05 %	0.038	0.006	83.01 %	158.410	34.429	78.27 %
8	0.375	0.053	85.87 %	0.046	0.006	86.24 %	193.408	25.767	86.68 %
9	0.428	0.064	84.98 %	0.050	0.006	87.85 %	207.477	42.404	79.56 %
10	0.477	0.057	87.95 %	0.057	0.006	89.14 %	228.735	19.131	91.64 %
Nos.	Recommendation			Letter			Chess		
	Alg-1	Alg-2	PDCT	Alg-1	Alg-2	PDCT	Alg-1	Alg-2	PDCT
1	86.511	84.897	1.87 %	168.543	164.441	2.43 %	5.250	5.002	4.73 %
2	132.889	57.667	56.61 %	290.299	137.046	52.79 %	8.921	5.643	36.74 %
3	204.054	78.257	61.65 %	414.714	157.428	62.04 %	13.492	6.578	51.24 %
4	252.085	40.780	83.82 %	539.906	177.505	67.12 %	17.120	6.380	62.73 %
5	326.254	61.032	81.29 %	669.274	196.077	70.70 %	22.334	8.460	62.12 %
6	375.863	42.035	88.82 %	733.782	128.112	82.54 %	28.909	9.093	68.55 %
7	451.967	62.027	86.28 %	878.486	203.763	76.81 %	33.532	8.961	73.28 %
8	508.167	44.592	91.22 %	1016.301	223.921	77.97 %	39.641	11.701	70.48 %
9	585.126	135.425	76.86 %	1113.369	198.810	82.14 %	43.988	11.958	72.82 %
10	641.297	45.844	92.85 %	1177.010	69.221	94.12 %	52.932	13.504	74.49 %

Note: (1) Alg-1 stands for Algorithm 1; (2) Alg-2 stands for Algorithm 2; (3) PDCT stands for percentage of the decreased computing times of Algorithm 2 to Algorithm 1.

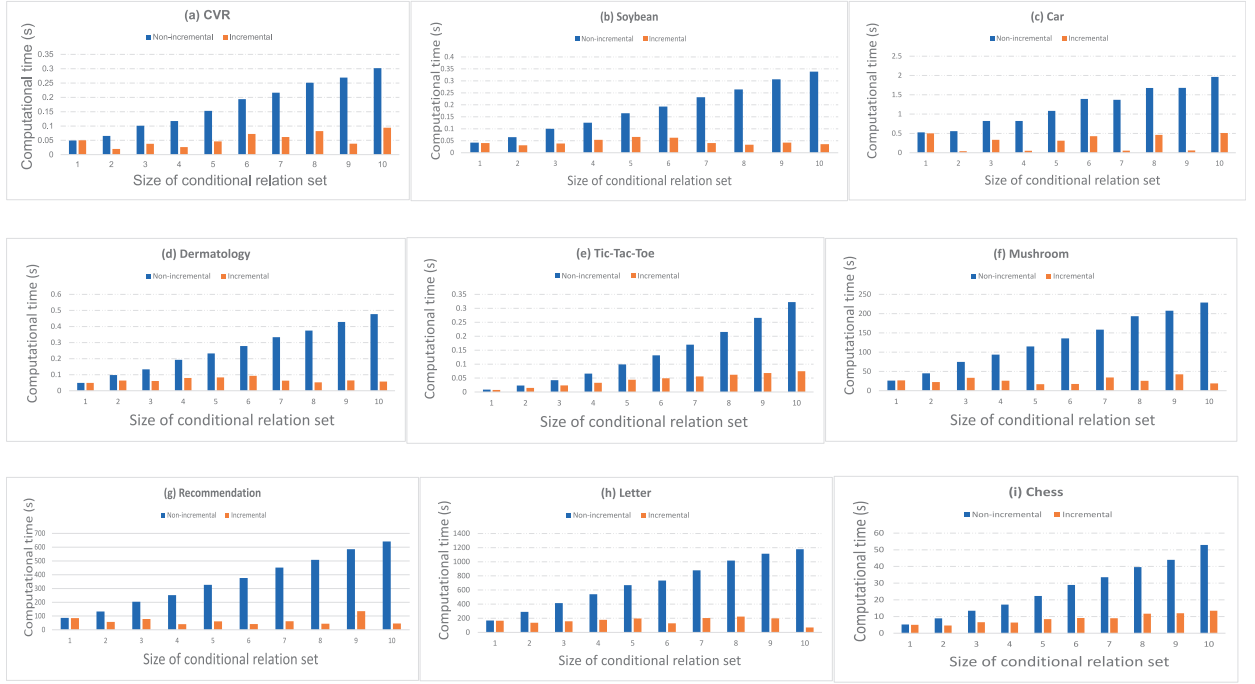


Fig. 5. Comparisons of computation times under non-incremental and incremental updating reduction algorithms over nine data sets when conditional relation set has increased.

As can be seen from the sub-graphs in Fig. 5 and Table 2, in the experiment on data set Nos. 1, the incremental updating algorithm of reduction has no obvious advantage in computational time compared with Algorithm 1 since there are not many historical computational results available. However, with the number of the conditional relations increases, it can be seen from column ‘PDCT’ in Table 2 that the computational time of the proposed incremental updating reduction algorithm is much smaller than that of the non-incremental updating reduction algorithm. At the same time, the differences of the computational times are also obvious in Fig. 5. This should be attributed to the fact that the proposed incremental updating reduction algorithm uses the historical computation results to compute the new positive region and avoids recomputing. Moreover, as the number of added conditional relations increases, the differences in computation times between the two algorithms become more pronounced. Therefore, the incremental updating reduction algorithm performs better in computation time when the conditional relation set has increased in a relation decision system.

5.2. Comparison of non-incremental and incremental updating reduction algorithms when conditional relation set has decreased

To show the performance of the proposed incremental updating algorithm of reduction (Algorithm 3), we compared non-incremental updating reduction algorithm (Algorithm 1) and Algorithm 3 on the nine datasets (in Table 1) when the conditional relation set has decreased. To distinguish the computational efficiency of Algorithm 3 on datasets with different numbers of conditional relation sets, we divide the conditional relation set of each dataset into 10 equal parts. Then remove the first part from the dataset resulting in the first data set (Nos. 1), remove the second part from the first data set resulting in the second data set (Nos. 2), and so on, until the ninth part is removed and the ninth data set (Nos. 9) is obtained. Since the conditional relation set is an empty set when the tenth segment data set is subtracted, such comparison is invalid, so no comparison is made in this segment. Obviously, the size of conditional relation set is gradually decreased by 10 %. Table 3 illustrates the comparison of the computation time of Algorithm 1 and Algorithm 3 in each data set (Nos. 1–Nos. 9). In each sub-graphs of Fig. 6, the x-coordinate represents the size of the conditional relation, and the y-coordinate represents the computation time.

As can be seen from the sub-graphs of Fig. 6 and Table 3, the computational time of the proposed incremental updating reduction algorithm is much less than that of the non-incremental updating reduction algorithm when the conditional relation set has decreased. In addition, it can be seen from the column ‘PDCT’ in Table 3 that with the increase of removed conditional relations, the differences of computational times between the two algorithms become obvious, which can also be seen in Fig. 6. This should be attributed to the fact that the proposed incremental updating reduction algorithm updates the new positive region based on the original positive region, which avoids many recomputing. Therefore, the proposed incremental updating reduction algorithm performs better in computational time when the conditional relation set has decreased in a relation decision system.

Table 3

Comparisons of computation times under non-incremental and incremental updating reduction algorithms over nine data sets when conditional relation set has decreased.

Nos.	CVR			Soybean			Car		
	Alg-1	Alg-3	PDCT	Alg-1	Alg-3	PDCT	Alg-1	Alg-3	PDCT
1	0.267	0.074	72.32 %	0.305	0.108	64.72 %	1.672	0.301	81.99 %
2	0.244	0.063	74.25 %	0.264	0.089	66.24 %	1.664	0.253	84.76 %
3	0.210	0.051	75.68 %	0.233	0.077	67.19 %	1.373	0.200	85.46 %
4	0.192	0.045	76.32 %	0.198	0.075	62.10 %	1.370	0.192	86.00 %
5	0.156	0.034	78.15 %	0.166	0.051	69.28 %	1.085	0.144	86.69 %
6	0.117	0.004	96.60 %	0.126	0.037	70.34 %	0.807	0.089	88.95 %
7	0.103	0.003	97.24 %	0.097	0.025	74.35 %	0.808	0.085	89.43 %
8	0.066	0.003	96.02 %	0.066	0.011	83.94 %	0.521	0.042	92.01 %
9	0.048	0.002	95.58 %	0.040	0.002	95.29 %	0.510	0.013	97.48 %
Nos.	Dermatology			Tic-Tac-Toe			Mushroom		
	Alg-1	Alg-3	PDCT	Alg-1	Alg-3	PDCT	Alg-1	Alg-3	PDCT
1	0.436	0.177	59.46 %	0.052	0.021	59.72 %	209.068	55.476	73.46 %
2	0.375	0.141	62.52 %	0.045	0.019	58.44 %	191.392	48.603	74.61 %
3	0.330	0.122	63.04 %	0.041	0.016	61.30 %	158.879	38.917	75.51 %
4	0.264	0.098	63.02 %	0.032	0.013	58.86 %	137.473	32.625	76.27 %
5	0.225	0.083	63.14 %	0.030	0.013	57.09 %	115.765	1.422	98.77 %
6	0.191	0.067	64.87 %	0.024	0.010	59.47 %	96.065	20.679	78.47 %
7	0.134	0.046	65.99 %	0.020	0.007	65.20 %	76.049	15.660	79.41 %
8	0.097	0.031	68.03 %	0.013	0.005	63.94 %	46.535	8.743	81.21 %
9	0.047	0.014	70.71 %	0.008	0.003	64.58 %	27.351	3.800	86.11 %
Nos.	Recommendation			Letter			Chess		
	Alg-1	Alg-3	PDCT	Alg-1	Alg-3	PDCT	Alg-1	Alg-3	PDCT
1	586.563	4.192	99.29 %	1192.496	255.515	78.57 %	43.033	13.351	68.97 %
2	509.670	133.280	73.85 %	1068.469	238.636	77.67 %	37.691	11.114	70.51 %
3	456.890	115.949	74.62 %	918.744	199.418	78.29 %	34.536	8.983	73.99 %
4	375.316	3.800	98.99 %	778.070	160.266	79.40 %	27.769	7.312	73.67 %
5	327.435	3.665	98.88 %	709.496	139.304	80.37 %	22.310	5.480	75.43 %
6	252.401	3.459	98.63 %	575.986	107.110	81.40 %	17.779	3.614	79.67 %
7	204.924	3.268	98.41 %	434.911	68.915	84.15 %	13.987	2.446	82.52 %
8	134.000	28.082	79.04 %	301.804	36.240	87.99 %	8.704	1.358	84.40 %
9	86.881	16.462	81.05 %	177.284	14.257	91.96 %	5.220	0.654	87.48 %

Note: (1) Alg-1 stands for Algorithm 1; (2) Alg-3 stands for Algorithm 3; (3) PDCT stands for percentage of the decreased computing times of Algorithm 3 to Algorithm 1.

5.3. Comparison on classification accuracy

To evaluate the quality of the reduction produced by the proposed incremental updating reduction algorithms (Algorithms 2, 3) when compared with the non-incremental updating reduction algorithm (Algorithm 1), we conduct the comparison of the classification accuracies derived by different algorithms and expressed in the form of mean value and standard deviation. Decision tree learning algorithm CART [2] is employed to train classifiers and test classification accuracies based on 10-fold cross validation. For each dataset in Table 1, we first divided the dataset into 10 subsets of almost equal size, of which 9 subsets are served as the training set and the remaining acts as the test set. Classification accuracies are achieved by reduction engendered by the proposed algorithms of which Algorithms 1 and 2 are employed to calculate the reductions when 80 % of whole conditional relation set is served as the original conditional relation set and the other 20 % as the increased conditional relation set. Meanwhile, Algorithms 1 and 3 are employed to calculate the reductions when 20 % of conditional relation are randomly removed from original conditional relation set. The results of these experiments are summarized in Table 4, where the means and standard deviations are expressed as percentages and “Original” depicts the classification accuracies with the original data, namely, the classification accuracy of the original data under classifier CART.

As can be clearly seen from Table 4, the classification accuracies of the reduced dataset derived by Algorithm 1 are almost identical to those of the original dataset except for Soybean and Dermatology datasets. The main reason is that the size of these datasets are small, and over-reduction may lead to low classification accuracy. In addition, when adding some conditional relations into a decision system, the classification accuracies of the reduced dataset derived by Algorithm 2 are close to those of Algorithm 1 on nine datasets. The experimental results indicate that the proposed incremental reduction algorithm can obtain some viable reductions in a shorter running time, which is consistent with the results of our study. Similarly, when removing some conditional relations from a decision system, the classification accuracies of the reduced data sets obtained by Algorithm 3 are identical to those of Algorithm 1 on nine datasets. Some viable reductions are acquired by Algorithm 3 in a smaller running time. Compared with the non-algorithm, shorten the time to find the reduction without reducing the classification accuracy of these datasets.

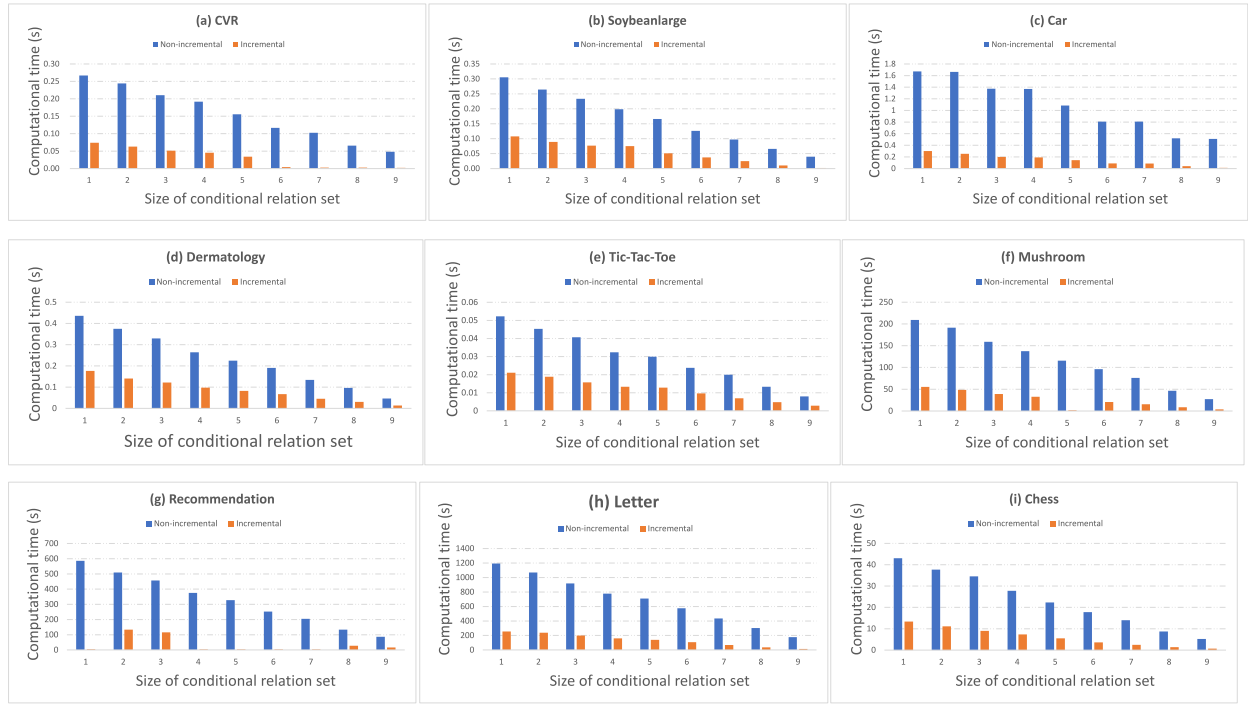


Fig. 6. Comparisons of computation times under non-incremental and incremental updating reduction algorithms over nine data sets when conditional relation set has decreased.

Table 4

Comparison of the classification accuracies of reduced data derived by Algorithm 1, 2 and 3 with CART classifier.

Data Set	Original	Adding conditional relation		Removing conditional relation	
		Algorithm 1	Algorithm 2	Algorithm 1	Algorithm 3
CVR	95.60 ± 2.50	95.38 ± 2.13	95.63 ± 2.50	96.34 ± 2.37	95.37 ± 2.92
Soybean	83.85 ± 5.63	73.23 ± 3.74	74.54 ± 3.85	74.56 ± 2.76	74.22 ± 2.96
Car	95.36 ± 0.78	95.08 ± 2.08	94.49 ± 1.76	67.78 ± 3.37	67.67 ± 3.47
Dermatology	94.50 ± 2.91	75.56 ± 6.63	95.34 ± 2.22	94.50 ± 2.57	94.80 ± 3.93
Tic-Tac-Toe	84.19 ± 3.15	81.26 ± 4.97	81.04 ± 4.35	80.52 ± 2.94	81.77 ± 4.31
Mushroom	99.86 ± 0.12	100.00 ± 0.00	100.00 ± 0.00	99.80 ± 0.23	99.86 ± 0.14
Recommendation	99.84 ± 0.17	99.98 ± 0.05	99.98 ± 0.07	99.69 ± 0.13	99.70 ± 0.19
Letter	84.29 ± 0.70	84.07 ± 0.02	84.69 ± 1.23	84.43 ± 0.41	84.20 ± 0.77
Chess	99.31 ± 0.53	99.15 ± 0.53	98.93 ± 0.42	82.31 ± 2.52	83.19 ± 2.11

To summaries, from the experimental results above, it can be seen that the proposed incremental reduction algorithms can not only reduce the time consumption of obtaining reduction, but also keep the classification performance basically unchanged.

6. Conclusions

In many tasks in the real world, decision systems may be dynamic, which can be regarded as dynamic relation decision systems. In dynamic relation decision systems, incremental update of reduction is helpful to improve the efficiency of knowledge discovery. In this paper, the incremental updating reduction algorithms are developed in relation decision systems with dynamic conditional relation sets. Firstly, the positive region incremental updating mechanism is established by the update method of right neighborhood, and then applied to the reduction calculation of a relation decision system with dynamic conditional relation sets. Finally, based on position region-based reduction method, the incremental updating reduction algorithms are designed for relation decision systems with dynamic conditional relation sets. Through a series of experiments, the performances of the proposed algorithms are evaluated. The results show that the proposed algorithm can effectively reduce the computational time of updating reduction in a relation decision system with dynamic conditional relation set.

Our future work will focus on:

- (a) Investigate how to update the reduction of relation decision systems with dynamic object sets and generalize the proposed method to practical applications;
- (b) Study how to update the reduction of relation decision system with relation change, and applies the proposed algorithm to practice;
- (c) Study how to update the reduction of a fuzzy relation decision systems and generalize the proposed approach to practical applications;
- (d) For a dynamic relation decision system, there may be multiple reductions. In application, from the viewpoint of optimization, the reductions found by the proposed algorithms may not be optimal. Therefore, it will be an interesting job to study an effective discernibility matrix updating mechanism, which can quickly find all the reductions.

CRedit authorship contribution statement

Lirun Su: Methodology, Writing – original draft. **Fusheng Yu:** Conceptualization, Supervision, Writing – review & editing. **Jinjin Li:** Conceptualization, Writing – review & editing. **Xubo Du:** Visualization. **Hanliang Huang:** Conceptualization, Writing – review & editing.

Data availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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